300Ø TO 400Ø	150×75 M.S CHANNEL	100×50 M.S CHANNEL
450Ø TO 600Ø	200×75 M.S CHANNEL	200×75 M.S CHANNEL

28.0 Section 23 07 19 - Insulation Pipe

1.0 GENERAL

The scope of this section comprises the supply and application of insulation conforming to these specifications.

1.01 SECTION INCLUDES

A. Piping insulation.

B. Jackets and accessories

1.02 RELATED REQUIREMENTS

A. Section 01 6116 - Volatile Organic Compound (VOC) Content Restrictions

B. Section 07 8400 - Firestopping.

C. Section 09 9000 - Painting and Coating: Painting insulation jacket.

D. Section 22 1005 - Plumbing Piping: Placement of hangers and hanger inserts.

E. Section 23 2113 - Hydronic Piping: Placement of hangers and hanger inserts.

F. Section 23 2300 - Refrigerant Piping: Placement of inserts.

1.03 REFERENCE STANDARDS

A.	ASHRAE (FUND)	ASHRAE Handbook - Fundamentals; latest version.
B.	ASTM A653/A653M	Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2011.
C.	ASTM B209M	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate [Metric]; 2010.
D.	ASTM E84	Standard Test Method for Surface Burning Characteristics of Building Materials; 2012.
E.	ASTM G21 - 2009	Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi
F.	ASTM E96/E96M-2010	Standard Test Methods for Water Vapor Transmission of Materials
G.	ASTM C612 -2010	Standard Specification for Mineral Fiber Block and Board Thermal Insulation
H.	ASTM E84 -2012	Standard Test Method for Surface Burning Characteristics of Building Materials
I.	ASTM C1338 -2008	Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings

Noted

J.	ASTM C518 -2010	Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus
K.	ASTM C553 -2011	Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications
L.	ASTM B209M - 2010	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate [Metric]
M.	NFPA 90A	Standard for the Installation of Air-Conditioning and Ventilating Systems; National Fire Protection Association; 2012.
N.	NFPA 255	Standard Method of Test of Surface Burning Characteristics of Building Materials
O.	NFPA 90B	Standard for the Installation of Warm Air Heating and Air Conditioning Systems; National Fire Protection Association; 2012.
P.	NFPA 96	Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations; National Fire Protection Association; 2011
Q.	SMACNA (KVS)	Kitchen Ventilation Systems and Food Service Equipment Fabrication & Installation Guidelines; 2001.
R.	UL 723	Standard for Test for Surface Burning Characteristics of Building Materials
S.	NBC	National Building Code of India (NBC) 2016
T.	BS 874 part 2 – 86 (DIN 52613 52612	Thermal Insulation Testing's; Determination of Thermal Conductivity by The Tube Method
U.	DIN EN 12667	Thermal performance of building materials and products - Determination of thermal resistance by means of guarded hot plate and heat flow meter methods - Products of high and medium thermal resistance
V.	DIN 53122-part 2	**Determination of the water vapor transmission rate of plastic film, rubber sheeting, paper, board and other sheet materials by gravimetry**
W.	DIN 52615	Test for Water Vapor Permeability
X.	EN 12086	Thermal Insulating Products for Building Applications - Determination of Water Vapor Transmission Properties
Y.	ASTM2180	Standard Test Method for Active Oxygen in Bleaching Compounds
Z.	ISO 5659	Plastics - Smoke generation - Part 2: Determination of optical density by a single-chamber test
AA.	DIN 52616	Testing of Thermal Insulation Materials; Determination of Thermal Conductivity by Means of a Heat-Flow Meter
AB.	IS: 8225	**Measurement of sound absorption in a reverberation room**

Noted

AC	ISO 354	Measurement of sound absorption in a reverberation room
AD	ASTM423C	**Standard Test Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method**
AE	UL 94	**Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances**
AF	BS 476 Part 7	Fire Tests on Building materials and structures – Method of test to determine the classification of the surface spread of flame of product
AG	EN ISO8497	**Thermal insulation - Determination of steady-state thermal transmission properties of thermal insulation for circular pipes (ISO 8497:1994)**
AH	EN13469	Thermal insulating products for building equipment and industrial installations. Determination of water vapor transmission properties of preformed pipe insulation
AI	ASTM C177	Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded Hot Plate Apparatus; 2010
AJ	ASTM C195	Standard Specification for Mineral Fiber Thermal Insulating Cement; 2007.
AK	ASTM C449	Standard Specification for Mineral Fiber Hydraulic-Setting Thermal Insulating and Finishing Cement; 2007 (Reapproved 2013).
AL	ASTM C533	Standard Specification for Calcium Silicate Block and Pipe Thermal Insulation; 2013
AM	ASTM 534/C534M	Standard Specification for Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form; 2011.
AN	ASTM C547	Standard Specification for Mineral Fiber Pipe Insulation; 2012.
AO	ASTM C552	**Standard Specification for Cellular Glass Thermal Insulation; 2012.**
AP	ASTM C578	Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation; 2012
AQ	ASTM C585	**Standard Practice for Inner and Outer Diameters of Rigid Thermal Insulation for Nominal Sizes of Pipe and Tubing (NPS System); 2010.**
AR	ASTM C591	**Standard Specification for Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation; 2012a.**
AS	ASTM C610	Standard Specification for Molded Expanded Perlite Block and Pipe Thermal Insulation; 2011

Noted

AT	ASTM C795	Standard Specification for Thermal Insulation for Use in Contact with Austenitic Stainless Steel; 2008.
AU	ASTM D1056	Standard Specification for Flexible Cellular Materials--Sponge or Expanded Rubber; 2007.
AV	ASTM D2842	**Standard Test Method for Water Absorption of Rigid Cellular Plastics; 2012.**

1.04 SUBMITTALS

A. Refer Special condition of contract for - Administrative Requirements, for submittal procedures.

B. Product Data: Provide product description, thermal characteristics, list of materials and thickness for each service, and locations.

C. Samples: Submit two samples of any representative size illustrating each insulation type. Samples of insulation material from each lot delivered at site may be selected by Owner's site representative and gotten tested for thermal conductivity and density at Contractor's cost. Adhesive used for sealing the insulation shall be non-flammable and with low VOC content (maximum 850 gm/l as per IGBC guidelines) strictly as per manufacturer's recommendations.

D. Manufacturer's Instructions: Indicate installation procedures necessary to ensure acceptable workmanship and that installation standards will be achieved.

1.05 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing products of the type specified in this section with not less than ten years of documented experience.

B. Applicator Qualifications: Company specializing in performing the type of work specified in this section, with minimum ten years of experience and approved by manufacturer.

C. Each lot of insulation material delivered at site shall be accompanied with manufacturer's test certificate for thermal conductivity values, density, water vapor permeability and fire properties.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Accept materials on site in original factory packaging, labelled with manufacturer's identification, including product density and thickness.

B. Protect insulation from weather and construction traffic, dirt, water, chemical, and mechanical damage, by storing in original wrapping.

C. Adhesives should be stored in cool and dry places not exposed to atmosphere.

1.07 FIELD CONDITIONS

A. Maintain ambient temperatures and conditions required by manufacturers of adhesives, mastics, and insulation cements.

B. Maintain temperature during and after installation for minimum period of 24 hours.

2.0 PRODUCTS

2.01 REQUIREMENTS FOR ALL PRODUCTS OF THIS SECTION

A. The insulation material shall be Closed Cell Elastomeric Nitrile Rubber **/ Polyethylene Foam / EPDM**

B. The scope of this section comprises the supply and application of all type of insulation material confirming to these technical specifications

Noted

C. Insulating material in tube form (minimum upto 100 dia pipes) shall be sleeved on the pipes. On piping, slit opened tube from insulating material shall be placed over the pipe and adhesive shall be applied as suggested by the manufacturer. Adhesive must be allowed to tack dry and then press surface firmly together starting from butt end and working towards centre. Wherever flat sheets shall be used it shall be with self-adhesive and cut out in correct dimension using correct tools. Scissors or Hacksaw-blade shall not be allowed.

D. All longitudinal and transverse joints shall be sealed by providing 50 mm wide fiber glass cloth laminated tape as per manufacturer recommendations. The adhesive shall be strictly as recommended by the manufacturer. The insulation shall be continuous over the entire run of piping, fittings and valves. All valves, fittings, joints, strainers etc. in chilled water piping shall be insulated to the same thickness as specified for the main run of piping and application shall be same as above. Valves bonnet, yokes and spindles shall be insulated in such a manner as not to cause damage to insulation when the valve is used or serviced.

E. **Direct contact between pipe and hanger shall be avoided. Hangers shall pass outside the saddle. Manufacturer shall supply PUF saddles with pre-laminated insulation sheet of both side (PUF saddle sandwich between insulation material on both side) so that the insulation material is joint with insulation material on both side (only for Nitrile & EPDM) and the weight of pipe is transferred to the PUF saddle in the center.**

F. Manufacturer's installation manual shall be submitted and followed for full compliance. All insulation work shall be carried out by skilled workmen specially trained and certified by manufacturer in this kind of work. All insulated pipes shall be labeled (S.R. or R.R.) and provided with 300 mm wide band of paint along circumference at every 1200 mm for colour coding. Direction of fluid shall also be marked. Un-insulated MS pipes shall be painted throughout and direction of fluid marked, two coats of zinc chromate primer and two coats of enamel paint. All painting shall be as per relevant BIS codes.

Noted

2.02 OPTION -1 CLOSED CELL ELASTOMERIC NITRILE RUBBER/EPDM

A. Manufacturers: Refer Special Conditions of Contract (SCC).

B. Thermal insulation material for Pipe insulation shall be with factory laminated black fiber glass cloth closed cell Elastomeric UV resistant. Thermal conductivity as per Reference Standards and codes.

C. Thermal conductivity as per of the insulation material shall not exceed 0.038 W/m°K or 0.212 BTU / (Hr-ft^2-°F/inch) at an average temperature of 30°C.

D. Density of the nitrile rubber shall be 40-60 Kg/m^3, for EPDM shall be 40-60 Kg/m^3 & for polyethylene material it shall be 25-30 Kg/m^3.

E. The product shall have temperature range of -(minus) 40 °C to 105°C. The insulation material shall be fire rated for Class 0 as per Reference Standards and codes for fire propagation test and for Class 1 as per BS 476 Part 7, 1987 for surface spread of flame test.

F. Water vapor permeability shall be not less than 0.024 per inch (2.48 x 10-13 Kg/ms.Pa

i.e. µ>7000: Water vapor diffuse on resistance) as ASTM G-21 and bacterial resistance to ASTM2180. In addition to above properties the insulation material for ducts shall be anti-microbial. Microbiological growth on insulation surface and bacterial resistance shall be in accordance with Reference Standards and codes.

Comply

The Material shall comply to ISO 5659 / BS 6853 / ABD 0031 for smoke density and toxicity values.

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G. The thermal conductivity of insulation material shall not be affected by aging as per **DIN 52616 standard.**

H. Insulation shall be with self-adhesive for ducting and piping and available in roles / sheets.

I. Thickness of the insulation shall be as specified for the individual application. Each lot of insulation material delivered at site shall be accompanied with manufacturer's test certificate for density and thickness. Samples of insulation material from each lot delivered at site may be selected by Owner's site representative and gotten tested for thermal conductivity and density at Contractor's cost. Adhesive used for sealing the insulation shall be non-flammable and with low VOC content (maximum 850 gm/l as per IGBC guidelines) strictly as per manufacturer's recommendations.

Comply

2.03 OPTION -2 "TF" QUALITY EXPANDED POLYSTYRENE INSULATION

A. Manufacturers: Refer Special Conditions of Contract (SCC).

B. "TF" Quality Expanded Polystyrene

1. All chilled water, refrigerant, and condensate drain piping shall be insulated in the manner specified herein. Before applying insulation, all pipe work and fittings shall be brushed and cleaned, and dust, dirt, mortar, and oil removed. All MS pipes shall be provided with a coat of zinc chromate primer, followed by two coats of cold setting adhesive compound. Thermal insulation shall then be applied as per the Schedule of Materials.

2. Pre-molded pipe sections shall be placed over the pipes, the longitudinal and transversal joints of these pipe sections shall be sealed with the adhesive compound. The insulation shall be continuous over the entire run of piping, fittings and valves.

3. Insulation shall be applied only after the piping system has been satisfactorily tested for leaks at 2 times the working pressure or at minimum 10 kg/sq.cm. test pressure.

4. All insulated pipes shall be covered with two layers of 400 gage polythene sheet to act as vapor barrier. PVC straps at 400 mm center shall be used to hold insulation and vapor barrier together.

5. Insulation shall be covered with 28-gauge aluminum cladding as per Schedule of Quantities and finished in neat and clean manner so as to achieve true surface.

6. All longitudinal and transverse joints in the outer cladding shall have a minimum overlap of 50 mm duly beaded and grooved and shall be sealed with elastomeric metal sealant 95-44 of Benjamin Foster USA, or equivalent.

7. Use of screws for fastening may puncture vapor barrier hence GI bands 0.50mm thick x 25 mm wide shall be provided at every 500 mm to retain cladding in position. Adhesive component once opened shall be used immediately and no leftovers shall be permitted to be used following day.

8. All insulation work shall be carried out by skilled workmen specially trained in this kind of work.

Noted

9. All insulated pipes shall be labeled (S, R or RR) and provided with band of paint for color coding as per IS codes. Direction of fluid shall also be marked.

10. Un-insulated M S pipes shall be painted throughout, and direction of fluid marked. All painting shall be as per relevant BIS codes.

11. Insulated pipes exposed to UV rays shall be covered with fiberglass fabric. Over fabric one coat of fire proof or Epoxy acrylic compound shall be applied. The coat shall be allowed to cure to non – stick state. Subsequently second coat of compound shall be applied to give tough and smooth finish to the insulated surface.

2.04 JACKETS

A. Canvas Jacket: UL listed 220 g/sq m plain weave cotton fabric treated with dilute fire-retardant lagging adhesive.

1. Lagging Adhesive:

a. Manufacturers: See approved makes List.

b. Compatible with insulation.

B. Mineral Fiber (Outdoor) Jacket: Asphalt impregnated and coated sheet, 50 lb/square (2.45 kg/sqm).

C. Aluminum Jacket: ASTM B209 (ASTM B209M).

1. Thickness: 0.016-inch (0.40 mm) sheet.

2. Finish: Smooth.

3. Joining: Longitudinal slip joints and 2-inch (50 mm) laps.

4. Fittings: 0.016 inch (0.4 mm) thick die shaped fitting covers with factory attached protective liner.

5. Metal Jacket Bands: 3/8 inch (10 mm) wide; 0.015 inch (0.38 mm) thick aluminum.

6. Metal Jacket Bands: 3/8 inch (10 mm) wide; 0.010 inch (0.25 mm) thick stainless steel.

2.05 Protective Coating / Vapor Barrier Over Insulation

All Pipes (On the roof / outside) exposed to UV rays shall be covered with two coats of UV paint / epoxy.

3.0 EXECUTION

3.01 EXAMINATION

A. Verify that Pipes have been tested before applying insulation materials.

B. Verify that surfaces are clean, foreign material removed, and dry.

3.02 INSTALLATION

A. Install in accordance with manufacturer's instructions.

B. Install in accordance with NAIMA (North American Insulation Manufacturers Association) / National Insulation Standards.

C. Insulated pipes conveying air below ambient temperature:

1. Provide insulation with vapor barrier jackets.

2. Finish with tape and vapor barrier jacket.

3. Continue insulation through walls, sleeves, hangers, and other pipes penetrations.

4. Insulate entire system including fittings, joints, flanges, fire dampers, flexible connections, and expansion joints.

Noted

5. Pre-molded pipe sections shall be placed over the pipes, the longitudinal and

transversal joints of these pipe sections shall be sealed with the adhesive compound. The insulation shall be continuous over the entire run of piping, fittings and valves.

6. Insulation shall be applied only after the piping system has been satisfactorily tested for leaks at 2 times the working pressure or at minimum 10 kg/sq.cm. test pressure.
7. All insulated pipes shall be covered with two layers of 400 gage polythene sheet to act as vapor barrier. PVC straps at 400 mm center shall be used to hold insulation and vapor barrier together.
8. Insulation shall be covered with 26-gauge GI sheet cladding as per Schedule of Quantities and finished in neat and clean manner so as to achieve true surface.
9. All longitudinal and transverse joints in the outer cladding shall have a minimum overlap of 50 mm duly beaded and grooved and shall be sealed with elastomeric metal sealant 95-44 of Benjamin Foster USA, or equivalent.
10. Use of screws for fastening may puncture vapor barrier hence GI bands 0.50mm thick x 25 mm wide shall be provided at every 500 mm to retain cladding in position. Adhesive component once opened shall be used immediately and no leftovers shall be permitted to be used following day.
11. All insulation work shall be carried out by skilled workmen specially trained in this kind of work.
12. All insulated pipes shall be labeled (S, R or RR) and provided with band of paint for color coding as per IS codes. Direction of fluid shall also be marked.
13. Un-insulated M S pipes shall be painted throughout, and direction of fluid marked. All painting shall be as per relevant BIS codes.
14. Insulated pipes exposed to UV rays shall be covered with fiberglass fabric. Over fabric one coat of fire proof or Epoxy acrylic compound shall be applied. The coat shall be allowed to cure to non – stick state. Subsequently second coat of compound shall be applied to give tough and smooth finish to the insulated surface.
15. Where Pipes penetrate walls / floor it shall be insulated with intumescent properties insulation material for fire protection. The treatment shall be minimum 500 mm extended on both sides

3.03 SCHEDULE OF MATERIAL

A. OPTION -1 CLOSED CELL ELASTOMERIC NITRILE RUBBER/EPDM

Thermal insulation shall be applied as follows or as specified in drawings or schedule of quantity:

Pipe nominal bore	Thickness for non-coastal places	Thickness for coastal places
15 mm – 25 mm	19 mm	25 mm
32 mm – 80 mm	25 mm	32 mm
100 mm – 400 mm	32 mm	38 mm
Above 400 mm	45 mm	45mm

Noted

B. OPTION - 2 TF" Quality Expanded Polystyrene

Thermal insulation shall be applied as follows or as specified in drawings or schedule

of quantity:

Pipe size (mm)	Thickness for T F Quality expanded polystyrene mm
10 to 40	25
50 to 150	50
150 to 450	75
Above 450	100

3.04 MODE OF MEASUREMENTS

Unless otherwise specified measurement for pipe insulation for the project shall be based on Centre line measurements described herewith

Pipe Insulation shall be measured in units of length along the center line of the installed pipe, strictly on the same basis as the piping measurements described earlier. The linear measurements shall be taken before the application of the insulation. It may be noted that for piping measurement, all valves, orifice plates and strainers are not separately measurable by their number and size. It is to be clearly understood that for the insulation measurements, all these accessories including cladding, valves, orifice plates and strainers shall be considered strictly by linear measurements along the center line of pipes and no special rate shall be applicable for insulation of any accessories, fixtures or fittings whatsoever. BIS Standards shall not be applicable for mode of measurements.

Noted

3.05 DATA SHEETS – APPENDIX "INSULATION – PIPE"

29.0 Section 23 07 13- Insulation Duct

2.0 GENERAL

The scope of this section comprises the supply and application of insulation conforming to these specifications.

1.01 SECTION INCLUDES

A. Duct insulation.

B. Duct Liner.

C. Insulation jackets.

1.02 RELATED REQUIREMENTS

A. Section 01 6116 - Volatile Organic Compound (VOC) Content Restrictions.

B. Section 07 8400 – Fire-stopping.

C. Section 09 9000 - Painting and Coating: Painting insulation jackets.

D. Section 23 3100 - HVAC Ducts and Casings.

1.03 REFERENCE STANDARDS

A.	ASHRAE (FUND)	ASHRAE Handbook - Fundamentals; latest version.
B.	ASTM A653/A653M	Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2011.
C.	ASTM B209M	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate [Metric]; 2010.
D.	ASTM E84	Standard Test Method for Surface Burning Characteristics of Building Materials; 2012.
E.	ASTM G21 - 2009	Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi
F.	ASTM E96/E96M-2010	Standard Test Methods for Water Vapor Transmission of Materials
G.	ASTM C612 -2010	Standard Specification for Mineral Fiber Block and Board Thermal Insulation
H.	ASTM E84 -2012	Standard Test Method for Surface Burning Characteristics of Building Materials
I.	ASTM C1338 -2008	Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings

Noted

J.	ASTM C1290 -2011	Standard Specification for Flexible Fibrous Glass Blanket Insulation Used to Externally Insulate HVAC Ducts
K.	ASTM C1071 2012(section 12.7).	Standard Specification for Fibrous Glass Duct Lining Insulation (Thermal and Sound Absorbing Material)
L.	ASTM C518 -2010	Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus
M.	ASTM C916 -2007	Standard Specification for Adhesives for Duct Thermal Insulation
N.	ASTM C553 -2011	Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications
O.	ASTM B209M -2010	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate [Metric]
P.	NFPA 90A	Standard for the Installation of Air-Conditioning and Ventilating Systems; National Fire Protection Association; 2012.
Q.	NFPA 255	Standard Method of Test of Surface Burning Characteristics of Building Materials
R.	NFPA 90B	Standard for the Installation of Warm Air Heating and Air Conditioning Systems; National Fire Protection Association; 2012.
S.	NFPA 96	Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations; National Fire Protection Association; 2011
T.	SMACNA (LEAK)	HVAC Air Duct Leakage Test Manual; Sheet Metal and Air Conditioning Contractors' National Association; 2012, 2nd Edition.
U.	SMACNA (DCS)	HVAC Duct Construction Standards.
V.	SMACNA (FGD)	Fibrous Glass Duct Construction Standards; Sheet Metal and Air Conditioning Contractors' National Association; 2003.
W.	SMACNA (KVS)	Kitchen Ventilation Systems and Food Service Equipment Fabrication & Installation Guidelines; 2001.
X.	UL 181	Standard for Factory-Made Air Ducts and Air Connectors; Underwriters Laboratories Inc.; Current Edition, Including All Revisions.
Y.	UL 1978	Grease Ducts; Current Edition, Including All Revisions
Z.	UL 723	Standard for Test for Surface Burning Characteristics of Building Materials
AA.	UL 2221	Tests of Fire Resistive Grease Duct Enclosure Assemblies; Current Edition, Including All Revisions.
AB.	NBC	National Building Code of India (NBC) latest version

Noted

AC	BS 874 part 2 – 86 (DIN 52613 52612	Thermal Insulation Testing's; Determination of Thermal Conductivity By The Tube Method
AD	DIN EN 12667	Thermal performance of building materials and products - Determination of thermal resistance by means of guarded hot plate and heat flow meter methods - Products of high and medium thermal resistance
AE	DIN 53122-part 2	Determination of the water vapor transmission rate of plastic film, rubber sheeting, paper, board, and other sheet materials by gravimetry
AF	DIN 52615	Test for Water Vapor Permeability
AG	EN 12086	Thermal Insulating Products for Building Applications - Determination of Water Vapor Transmission Properties
AH	ASTM2180	Standard Test Method for Active Oxygen in Bleaching Compounds
AI	ISO 5659	Plastics - Smoke generation - Part 2: Determination of optical density by a single-chamber test
AJ	BS 6853	Code of practice for fire precautions in the design and construction of passenger carrying trains
AK	ABD 0031	Airbus Fireworthiness Requirements Generation of smoke and toxic gases
AL	DIN 52616	Testing of Thermal Insulation Materials; Determination of Thermal Conductivity by Means of a Heat-Flow Meter
AM	IS: 8225	Measurement of sound absorption in a reverberation room
AN	ISO 354	Measurement of sound absorption in a reverberation room
AO	ASTM423C	Standard Test Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method
AP	UL 94	Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances
AQ	UL 1996	UL Standard for Safety Electric Duct Heaters
AR	BS 476 Part 6	Fire Tests on Building materials and structures – Method of test to determine the classification of the surface spread of flame of product

1.04 SUBMITTALS

A. Refer Special condition of contract for - Administrative Requirements, for submittal procedures.

Noted

.B. Product Data: Provide product description, thermal characteristics, list of materials and thickness for each service, and locations.

C. Samples: Submit two samples of any representative size illustrating each insulation type. Samples of insulation material from each lot delivered at site may be selected by Owner's site representative and gotten tested for thermal conductivity and density at Contractor's cost. Adhesive used for sealing the insulation shall be non-flammable and with low VOC content (maximum 850 gm/l as per IGBC guidelines) strictly as per manufacturer's recommendations.

D. Manufacturer's Instructions: Indicate installation procedures necessary to ensure acceptable workmanship and that installation standards will be achieved.

1.05 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing products of the type specified in this section with not less than ten years of documented experience.

B. Applicator Qualifications: Company specializing in performing the type of work specified in this section, with minimum ten years of experience and approved by manufacturer.

C. Each lot of insulation material delivered at site shall be accompanied with manufacturer's test certificate for thermal conductivity values, density, water vapor permeability and fire properties.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Accept materials on site in original factory packaging, labelled with manufacturer's identification, including product density and thickness.

B. Protect insulation from weather and construction traffic, dirt, water, chemical, and mechanical damage, by storing in original wrapping.

C. Adhesives should be stored in cool and dry places not exposed to atmosphere.

1.07 FIELD CONDITIONS

A. Maintain ambient temperatures and conditions required by manufacturers of adhesives, mastics, and insulation cements.

B. Maintain temperature during and after installation for minimum period of 24 hours.

2.0 PRODUCTS

2.01 REQUIREMENTS FOR ALL PRODUCTS OF THIS SECTION

A. The insulation material shall be factory laminated 7 mil thick black fiber glass cloth Closed Cell Elastomeric Nitrile Rubber UV resistant

B. Manufacturer::Refer Special Conditions of Contract (SCC)

C. The scope of this section comprises the supply and application of all type of insulation material confirming to these technical specifications.

D. Thermal insulation material for Duct insulation shall be with factory laminated black fiber glass cloth closed cell Elastomeric UV resistant.

E. Thermal conductivity as per of the insulation material shall not exceed 0.038 W/m^{o}K or 0.212 BTU / (Hr-ft^{2}-^{o}F/inch) at an average temperature of 30^{o}C.

F. Density of the nitrile rubber shall be 40-60 Kg/m^{3}

Noted

G. The product shall have temperature range of -(minus) 40 ^{o}C to 105^{o}C. The insulation material shall be fire rated for Class 0 as per Reference Standards and codes for fire propagation test and for Class 1 as per BS 476 Part 7, 1987 for surface spread of flame test.

H. Water vapor permeability shall be not less than 0.024 per inch (2.48 x 10-13 Kg/ms.Pa i.e. μ>7000: Water vapor diffuse on resistance) as per DIN 53122 part 2, DIN 52615 / EN

12086 & EN13469.

I. In addition to above properties the insulation material for ducts shall be anti-microbial for health care projects or wherever specified in the BOQ. Microbiological growth on insulation surface and bacterial resistance shall be in accordance with ASTMG-21 and bacterial resistance ASTM2180.

The Material shall comply to ISO 5659 / BS 6853 / ABD 0031 for smoke density and toxicity values.

J. The thermal conductivity of insulation material shall not be affected by aging as DIN 52616 standard.

k. Insulation shall be with self-adhesive for ducting and piping and available in roles / sheets.

L. Thickness of the insulation shall be as specified for the individual application. Each lot of insulation material delivered at site shall be accompanied with manufacturer's test certificate for density and thickness. Samples of insulation material from each lot delivered at site may be selected by Owner's site representative and gotten tested for thermal conductivity and density at Contractor's cost. Adhesive used for sealing the insulation shall be non-flammable and with low VOC content (maximum 850 gm/l as per IGBC guidelines) strictly as per manufacturer's recommendations.

2.02 JACKETS

A. Canvas Jacket: UL listed 220 g/sq m plain weave cotton fabric treated with dilute fire-retardant lagging adhesive.

1. Lagging Adhesive:
   a. Manufacturers: See SCC Appendix III for approved makes.
   b. Compatible with insulation.

B. Mineral Fiber (Outdoor) Jacket: Asphalt impregnated and coated sheet, 50 lb/square (2.45 kg/sq m).

C. Aluminum Jacket: ASTM B209 (ASTM B209M).

1. Thickness: 0.016-inch (0.40 mm) sheet.
2. Finish: Smooth.
3. Joining: Longitudinal slip joints and 2-inch (50 mm) laps.
4. Fittings: 0.016 inch (0.4 mm) thick die shaped fitting covers with factory attached protective liner.
5. Metal Jacket Bands: 3/8 inch (10 mm) wide; 0.015 inch (0.38 mm) thick aluminum.
6. Metal Jacket Bands: 3/8 inch (10 mm) wide; 0.010 inch (0.25 mm) thick stainless steel.

2.03 NITRILE RUBBER DUCT INSULATION

A. Manufacturers: Special Conditions of Contract (SCC).

B. Duct Acoustic Lining:

Noted

Open Cell Nitrile Rubber

Duct acoustic lining material shall be Nitrile Rubber open cell foam. Thermal conductivity of the insulation material shall not exceed 0.047 W/m°K at an average temperature of 20°C. Density of the nitrile rubber shall be 140 – 180 Kg/m^3. The material should withstand maximum surface temperature of +85°C and minimum surface temperature of

-20ºC. The material should conform to Class 1 rating for surface spread of Flame in accordance to BS 476 Part 7 & UL 94 (HBF, HF 1 & HF 2) in accordance to UL 94, 1996.

It should have antimicrobial product protection and should pass Fungi Resistance as per ASTM G 21 and Bacterial Resistance as per ASTM E 2180. The insulation should pass Air Erosion Resistance Test in accordance to ASTM Standard C 1071-05 (section 12.7).

Thickness of the material shall be 15 mm thick or as specified in the SOQ for the individual application. The insulation should be installed as per manufacturer's recommendation.

C. Acoustic lining in Plenums

Acoustic lining in Plenums especially for Air diffusion connected to slot diffusers shall have 12mm thick rigid board of fiberglass/mineral wool having density of 48 Kg/m3 / 15mm thick Nitrile rubber having density of 140 – 180 Kg/m3 or as specified in the SOQ.

D. Noise sensitive applications

For Noise sensitive applications such as Auditoriums, recording studios, etc. the acoustic insulation material shall have properties as tabled below:

		Absorption Coefficient at Octave band center Frequencies (Hz)						
Density	Thickness mm	125	250	500	1000	2000	4000	NRC
32 Kg/ m³	15	0.05	0.13	0.32	0.65	0.79	0.93	0.45
	25	0.29	0.45	0.77	1.0	0.93	0.96	0.8
48 Kg/ m³	15	0.05	0.12	0.29	0.51	0.68	0.8	0.4
	25	0.16	0.32	0.82	1.02	1.05	1.0	0.8
	50	0.3	0.85	1.03	1.07	1.06	1.0	1.0

Wherever acoustic consultant is involved in the project, the above table shall be vetted by him.

2.04 Protective Coating / Vapor Barrier Over Insulation

A. All ducts (On the roof / outside) exposed to UV rays shall be covered with two coats of UV paint / epoxy.

B.

2.05 Sound Attenuators

A. Attenuators shall be installed in ducts in accordance with requirements of drawings and as included in Schedule of Quantities

B. Noise levels
Within conditioned spaces Noise levels shall be not greater than those set out in schedule as per section 442000 – NOISE CONTROL

3.0 EXECUTION

3.01 EXAMINATION

Noted

A. Verify that ducts have been tested before applying insulation materials.

B. Verify that surfaces are clean, foreign material removed, and dry.

3.02 INSTALLATION

A. Install in accordance with manufacturer's instructions.

B. Install in accordance with NAIMA (North American Insulation Manufacturers Association) / National Insulation Standards.

C. Insulated ducts conveying air below ambient temperature:

1. Provide insulation with vapor barrier jackets.
2. Finish with tape and vapor barrier jacket.
3. Continue insulation through walls, sleeves, hangers, and other duct penetrations.
4. Insulate entire system including fittings, joints, flanges, fire dampers, flexible connections, and expansion joints.
5. The thickness of nitrile rubber shall be as shown on drawing or identified in the schedule of quantity. Following procedure shall be adhered to:

   The thickness of insulation material shall be as shown on drawings or identified in the schedule of quantity. Following procedure shall be adhered to:

   Duct surfaces shall be cleaned to remove all grease, oil, dirt, etc. prior to carrying out insulation work. Measurement of surface dimensions shall be taken properly to cut closed cell insulation to size with sufficient allowance in dimension. Cutting of insulation sheets shall be done with adjustable blade to make 90^0 cuts in thickness of sheet. Hacksaw or blades are not acceptable tools for cutting the insulation.

   Material shall be fitted under compression and no stretching of material shall be permitted. All longitudinal and transverse joints shall be sealed by providing 50 mm wide Fibre glass cloth laminated tape as per manufacturer recommendations. The insulation installers shall be certified by manufacture.

   Where ducts penetrate walls / floor it shall be insulated with intumescent properties insulation material for fire protection. The treatment shall be minimum 500 mm extended on both sides

D. Insulated ducts conveying air above ambient temperature:

1. Insulate fittings and joints. Where service access is required, bevel and seal ends of insulation.

E. Ducts Exposed in Mechanical Equipment Rooms or Finished Spaces (below 10 feet (3 meters) above finished floor): Finish with canvas jacket sized for finish painting.

F. Exterior Applications: Provide insulation with vapor barrier jacket. Cover with outdoor jacket finished as specified in Schedule of Quantities.

G. External Duct Insulation Application:

1. Secure insulation with vapor barrier with wires and seal jacket joints with vapor barrier adhesive or tape to match jacket.
2. Secure insulation without vapor barrier with staples, tape, or wires.
3. Install without sag on underside of duct. Use adhesive or mechanical fasteners where necessary to prevent sagging. Lift duct off trapeze hangers and insert spacers.

Noted

4. Seal vapor barrier penetrations by mechanical fasteners with vapor barrier adhesive.
5. Stop and point insulation around access doors and damper operators to allow operation without disturbing wrapping.

3.03 SCHEDULE OF MATERIAL

1. Ducting insulation thickness shall be as per table below.

Ducting position	Thickness for non-coastal places	Thickness for coastal places
SA duct in RA path	13 mm	**19 mm**
Ducted return air system	SA duct: 19 mm RA duct: 13 mm	**SA duct: 25 mm** **RA duct: 19 mm**
Both SA & RA exposed	Both 25 mm	**Both 25 mm**
SA duct having open cell nitrile rubber insulation	9 mm	**9 mm**

Anti-microbial duct insulation material thickness can be 1 mm less than corresponding standard nitrile rubber insulation

3.04 MODE OF MEASUREMENT

Unless otherwise specified measurement for duct insulation for the project shall be based on Centre line measurements described herewith:

Duct Insulation and Acoustic Lining shall be measured based on surface area along the Centre line of insulation thickness. Thus the surface area of externally thermally insulated or acoustically lined be based on the perimeter comprising Centre line (of thickness of insulation) width and depth of the cross section of insulated or lined duct, multiplied by the Centre-line length including tapered pieces, bends, tees, branches, etc. as measured for bare ducting. . BIS Mode of measurement shall not be applicable.

3.05 DATA SHEETS – APPENDIX “INSULATION – DUCTS”

30.0 Section 23 09 00.4 – Temperature Sensor

1.0 GENERAL

1.01 SECTION INCLUDES

A. Temperature sensor

1.02 RELATED REQUIREMENTS

A. Section 23 2113 - Hydronic Piping.

B. Section 23 2213 - Steam and Condensate Heating Piping.

C. Section 23 0943 - Pneumatic Control System for HVAC.

D. Section 23 0923 - Direct-Digital Control System for HVAC.

E. Section 23 0993 - Sequence of Operations for HVAC Controls.

1.03 REFERENCE STANDARDS

Noted

A.	IP 55	IP 55 Standard Protection Specification – Protection from dirt,